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SECOND PAGE: INTRODUCTION OF OUR GROUP

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Introduction

Vietnam has remained the world's leading exporter of cashew nuts for 18 consecutive years, accounting for over 80% of global export volume. In 2024 alone, the country exported more than 765,000 tons of cashew kernels, generating over 4.6 billion USD in revenue—a remarkable increase compared to previous years. Binh Phuoc Province, often referred to as the “cashew capital” of Vietnam, contributes nearly 50% of the national cultivation area and over 40% of the total raw cashew nut output.

Despite the industry's robust growth, large quantities of cashew testa—the thin, reddish-brown skin surrounding the kernel—are commonly discarded or used as low-value fuel or fertilizer. However, cashew testa is rich in bioactive compounds, including polyphenols, tannins, catechin, epicatechin, and gallic acid, which have been scientifically reported to exhibit antioxidant, anti-inflammatory, antimicrobial, and anti-allergic activities. This highlights the testa's potential as a valuable but underutilized resource in fields such as pharmaceuticals, cosmetics, and food technology.

This study aims to extract a polyphenol-rich fraction from cashew testa and evaluate its potential in the formulation of an antioxidant topical cream. Ultrasound-assisted extraction (UAE) was employed using various solvent systems to enhance extraction efficiency and bioactivity. Compared to conventional methods, UAE offers multiple advantages, including reduced processing time, broad solvent compatibility, and improved operational safety. Through this work, we seek to promote the valorization of agricultural by-products while contributing to sustainable development and the creation of high-value natural products.

Methods

1. Extraction of Cashew Testa Polyphenol-rich Extract

Collected cashew testa was pretreated to remove impurities and dust, then dried at 50°C and ground into fine powder. Extraction was performed using ultrasound-assisted extraction (UAE) with various solvents (ethanol, methanol, and water), under different conditions of time, temperature, and solid-to-solvent ratios. Extraction efficiency was evaluated based on yield and total polyphenol content (TPC), expressed as mg GAE/g extract.

2. Determination of Total Polyphenol Content (TPC)

TPC was determined using the Folin–Ciocalteu method, with results calculated from a standard calibration curve.

3. Evaluation of Biological Activities

3.1. Antioxidant Activity

Antioxidant capacity was assessed using DPPH and ABTS radical scavenging assays. Inhibition percentages and IC₅₀ values were determined.

3.2. Antibacterial Activity

The antibacterial effect of the extract was evaluated against *Staphylococcus aureus* ATCC 25923 and *Candida* spp, using disk diffusion and minimum inhibitory concentration (MIC) methods.

4. Cream Formulation

The cashew testa extract was incorporated into a topical cream formulation. The final product is a homogeneous, brown-colored cream, packed in 10g aluminum tubes, with smooth texture and no phase separation.

RESULTS

1. Extraction Yield of Cashew Testa Extracts

Different extraction solvents (ethanol and methanol, water) were applied using ultrasonic-assisted extraction. The results are summarized below:

Sample	Input mass (g)	Extract yield (g)	Extraction efficiency (%)
Ethanol	10	2.65	26.53 ^b ± 2.87
Methanol	10	3.61	36.07 ^a ± 1.7
Water	10	1.55	15.47 ^c ± 0.31

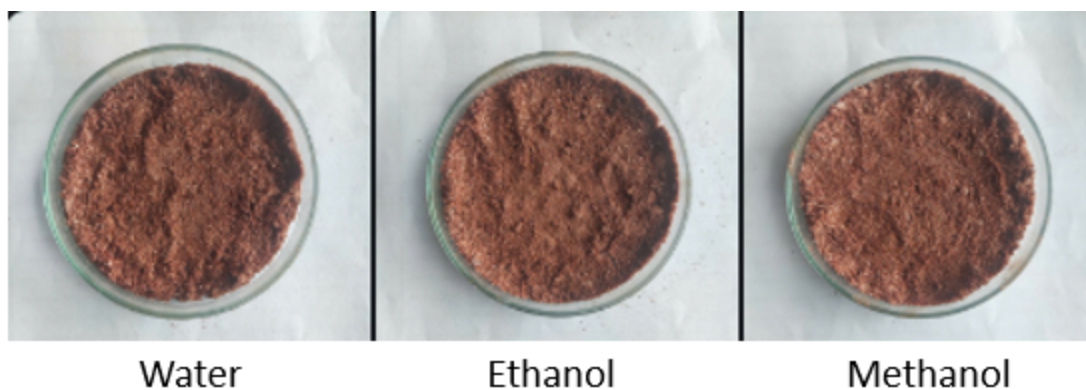


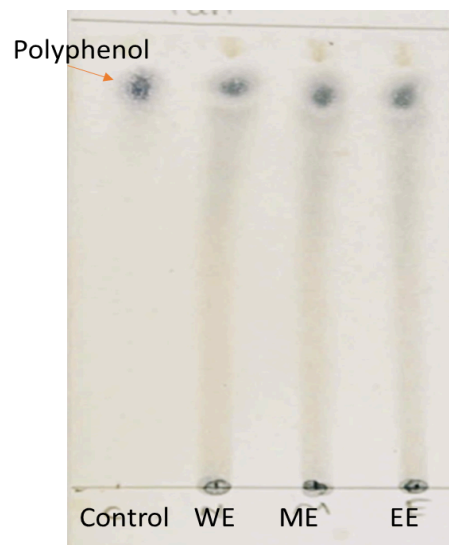
Figure: Photos of cashew testa extract powder from different extraction solvent

2. Phytochemical Screening and Polyphenol Content

Qualitative Results: The extract samples tested positive for polyphenols, tannins, and flavonoids

Sample	Total polyphenols (mg GAE/g extract)
Ethanol	465.33 ^b ± 4.69
Methanol	484.06 ^a ± 4.26
Water	453.13 ^c ± 0.65

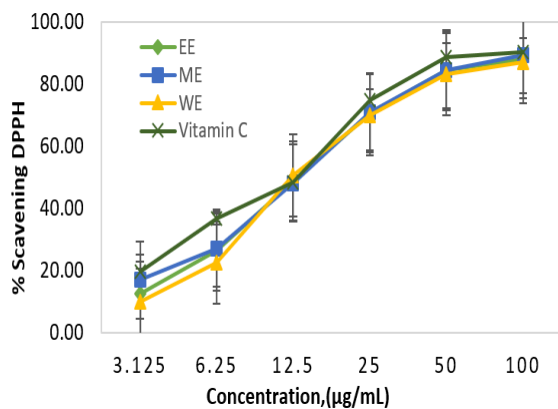
Sample	Tannin	Flavonoid	Polyphenol
Ethanol	+	+	+
Methanol	+	+	+
Water	+	+	+



Total Polyphenol Content:

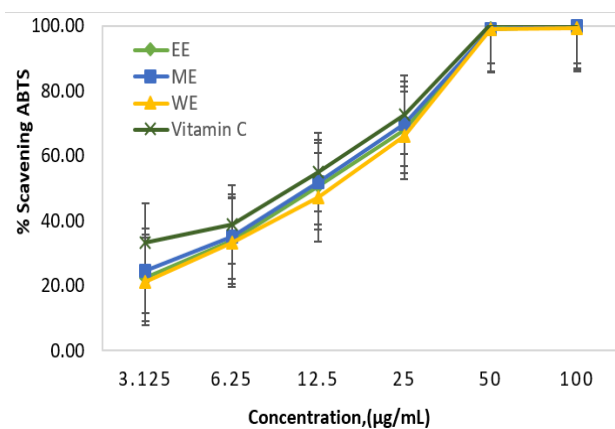
3. Antioxidant activity

DPPH Radical Scavenging activity (%):



Sample	Equation	IC50 value
Ethanol	$y=22.787\ln(x) - 9.674$ $R^2= 0.9686$	13.72
Methanol	$y=22.451\ln(x) - 7.7085$ $R^2= 0.9759$	13.07
Water	$y=23.913\ln(x) - 14.249$ $R^2= 0.9607$	14.68
Vitamin C	$y=22.05\ln(x) - 3.6337$ $R^2= 0.9603$	11.39

ABTS Radical Scavenging activity (%):



Sample	Equation	IC50 value
Ethanol	$y= 24.282\ln(x) - 6.9244$ $R^2= 0.9545$	10.43
Methanol	$Y = 23.646\ln(x) - 4.7629$ $R^2= 0.9462$	10.14
Water	$y= 24.956\ln(x) - 10.874$ $R^2= 0.9526$	11.36
Vitamin C	$y= 22.245\ln(x) + 2.8521$ $R^2= 0.9535$	8.32

IC50 values indicated the methanol extract has strong antioxidant capacity comparable to vitamin C.

4. Antibacterial Activity

Antimicrobial tests against *Staphylococcus aureus* ATCC 25923 and *Candida albicans* showed the ethanol extract exhibited inhibitory zones, demonstrating promising antibacterial potential.

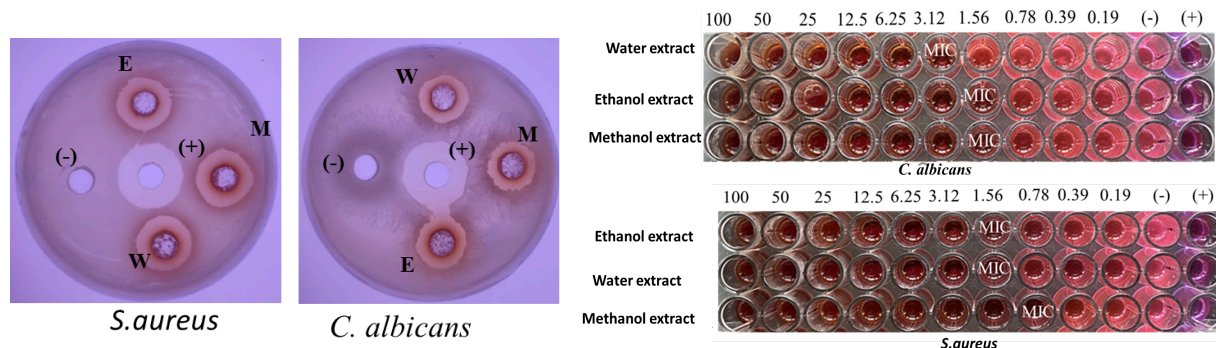


Figure: Images of inhibition zones under disk diffusion method and MIC against *Staphylococcus aureus* ATCC 25923 and *Candida albicans*

Sample	Diameter of inhibition zone of <i>S.aureus</i> (mm)	Diameter of inhibition zone of <i>C.albicans</i> (mm)
Ethanol	12,5 ^a ± 0,5	11,17 ^a ± 0,29
Methanol	13,5 ^a ± 0,5	11,33 ^a ± 0,58
Water	11,17 ^b ± 0,29	10,5 ^a ± 0,5

5. Cream Formulation

The cashew testa extract was incorporated into a topical cream formulation. The final product is a homogeneous, brown-colored cream, packed in 10g aluminum tubes, with smooth texture and no phase separation.

Figure:

Conclusion

- Cashew testa extract (especially ethanol-based) exhibited high polyphenol content with strong antioxidant and antimicrobial activities.

- A stable, natural cream was successfully formulated using the extract, demonstrating practical cosmetic application.
- The project promotes waste valorization and supports sustainable, eco-friendly innovation in agriculture and skincare.

Acknowledgments

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References

[List of main references or "Available upon request" depending on brochure format]